

18.1 Basic configuration

The two SSP interfaces, SSP0 and SSP1 are configured using the following registers:

1. Power: In the PCONP register ([Table 46](#)), set bit PCSSP0 to enable SSP0 and bit PCSSP1 to enable SSP1.
Remark: On reset, both SSP interfaces are enabled (PCSSP0/1 = 1).
2. Clock: In PCLKSEL0 select PCLK_SSP1; in PCLKSEL1 select PCLK_SSP0 (see [Section 4.7.3](#)). In master mode, the clock must be scaled down (see [Section 18.6.5](#)).
3. Pins: Select the SSP pins through the PINSEL registers ([Section 8.5](#)) and pin modes through the PINMODE registers ([Section 8.4](#)).
4. Interrupts: Interrupts are enabled in the SSP0IMSC register for SSP0 and SSP1IMSC register for SSP1 [Table 376](#). Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register, see [Table 50](#).
5. Initialization: There are two control registers for each of the SSP ports to be configured: SSP0CR0 and SSP0CR1 for SSP0, SSP1CR0 and SSP1CR1 for SSP1. See [Section 18.6.1](#) and [Section 18.6.2](#).
6. DMA: The Rx and Tx FIFOs of the SSP interfaces can be connected to the GPDMA controller (see [Section 18.6.10](#)). For GPDMA system connections, see [Table 544](#).

Remark: SSP0 is intended to be used as an alternative for the SPI interface, which is included as a legacy peripheral. Only one of these peripherals can be used at the any one time.

18.2 Features

- Compatible with Motorola SPI, 4-wire TI SSI, and National Semiconductor Microwire buses.
- Synchronous Serial Communication.
- Master or slave operation.
- 8 frame FIFOs for both transmit and receive.
- 4 to 16 bit data frame.
- DMA transfers supported by GPDMA.

18.3 Description

The SSP is a Synchronous Serial Port (SSP) controller capable of operation on a SPI, 4-wire SSI, or Microwire bus. It can interact with multiple masters and slaves on the bus. Only a single master and a single slave can communicate on the bus during a given data transfer. Data transfers are in principle full duplex, with frames of 4 to 16 bits of data flowing from the master to the slave and from the slave to the master. In practice it is often the case that only one of these data flows carries meaningful data.

The LPC176x/5x has two Synchronous Serial Port controllers -- SSP0 and SSP1.

18.4 Pin descriptions

Table 369. SSP pin descriptions

Pin Name	Type	Interface pin name/function			Pin Description
		SPI	SSI	Microwire	
SCK0/1	I/O	SCK	CLK	SK	Serial Clock. SCK/CLK/SK is a clock signal used to synchronize the transfer of data. It is driven by the master and received by the slave. When the SPI interface is used, the clock is programmable to be active-high or active-low, otherwise it is always active-high. SCK1 only switches during a data transfer. Any other time, the SSPn interface either holds it in its inactive state, or does not drive it (leaves it in high-impedance state).
SSEL0/1	I/O	SSEL	FS	CS	Frame Sync/Slave Select. When the SSPn interface is a bus master, it drives this signal to an active state before the start of serial data, and then releases it to an inactive state after the serial data has been sent. The active state of this signal can be high or low depending upon the selected bus and mode. When the SSPn is a bus slave, this signal qualifies the presence of data from the Master, according to the protocol in use. When there is just one bus master and one bus slave, the Frame Sync or Slave Select signal from the Master can be connected directly to the slave's corresponding input. When there is more than one slave on the bus, further qualification of their Frame Select/Slave Select inputs will typically be necessary to prevent more than one slave from responding to a transfer.
MISO0/1	I/O	MISO	DR(M) DX(S)	SI(M) SO(S)	Master In Slave Out. The MISO signal transfers serial data from the slave to the master. When the SSPn is a slave, serial data is output on this signal. When the SSPn is a master, it clocks in serial data from this signal. When the SSPn is a slave and is not selected by FS/SSEL, it does not drive this signal (leaves it in high-impedance state).
MOSI0/1	I/O	MOSI	DX(M) DR(S)	SO(M) SI(S)	Master Out Slave In. The MOSI signal transfers serial data from the master to the slave. When the SSPn is a master, it outputs serial data on this signal. When the SSPn is a slave, it clocks in serial data from this signal.

18.5 Bus description

18.5.1 Texas Instruments synchronous serial frame format

[Figure 77](#) shows the 4-wire Texas Instruments synchronous serial frame format supported by the SSP module.

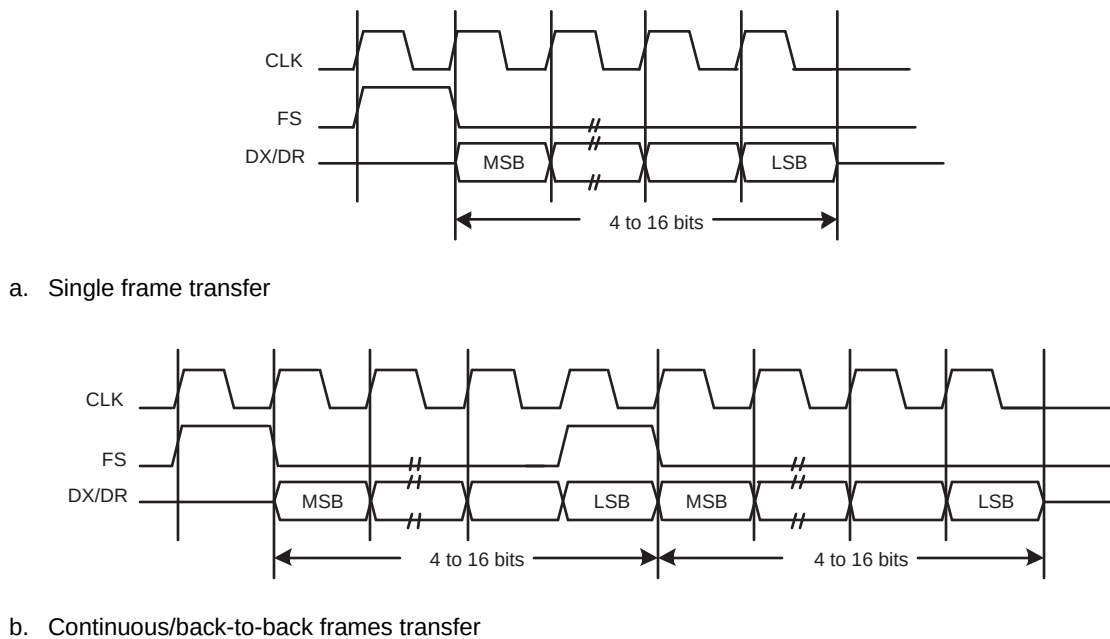


Fig 77. Texas Instruments Synchronous Serial Frame Format: a) Single and b) Continuous/back-to-back Two Frames Transfer

For device configured as a master in this mode, CLK and FS are forced LOW, and the transmit data line DX is tri-stated whenever the SSP is idle. Once the bottom entry of the transmit FIFO contains data, FS is pulsed HIGH for one CLK period. The value to be transmitted is also transferred from the transmit FIFO to the serial shift register of the transmit logic. On the next rising edge of CLK, the MSB of the 4-bit to 16-bit data frame is shifted out on the DX pin. Likewise, the MSB of the received data is shifted onto the DR pin by the off-chip serial slave device.

Both the SSP and the off-chip serial slave device then clock each data bit into their serial shifter on the falling edge of each CLK. The received data is transferred from the serial shifter to the receive FIFO on the first rising edge of CLK after the LSB has been latched.

18.5.2 SPI frame format

The SPI interface is a four-wire interface where the SSEL signal behaves as a slave select. The main feature of the SPI format is that the inactive state and phase of the SCK signal are programmable through the CPOL and CPHA bits within the SSPCR0 control register.

18.5.2.1 Clock Polarity (CPOL) and Phase (CPHA) control

When the CPOL clock polarity control bit is 0, it produces a steady state low value on the SCK pin. If the CPOL clock polarity control bit is 1, a steady state high value is placed on the CLK pin when data is not being transferred.

The CPHA control bit selects the clock edge that captures data and allows it to change state. It has the most impact on the first bit transmitted by either allowing or not allowing a clock transition before the first data capture edge. When the CPHA phase control bit is 0, data is captured on the first clock edge transition. If the CPHA clock phase control bit is 1, data is captured on the second clock edge transition.

18.5.2.2 SPI format with CPOL=0,CPHA=0

Single and continuous transmission signal sequences for SPI format with CPOL = 0, CPHA = 0 are shown in [Figure 78](#).

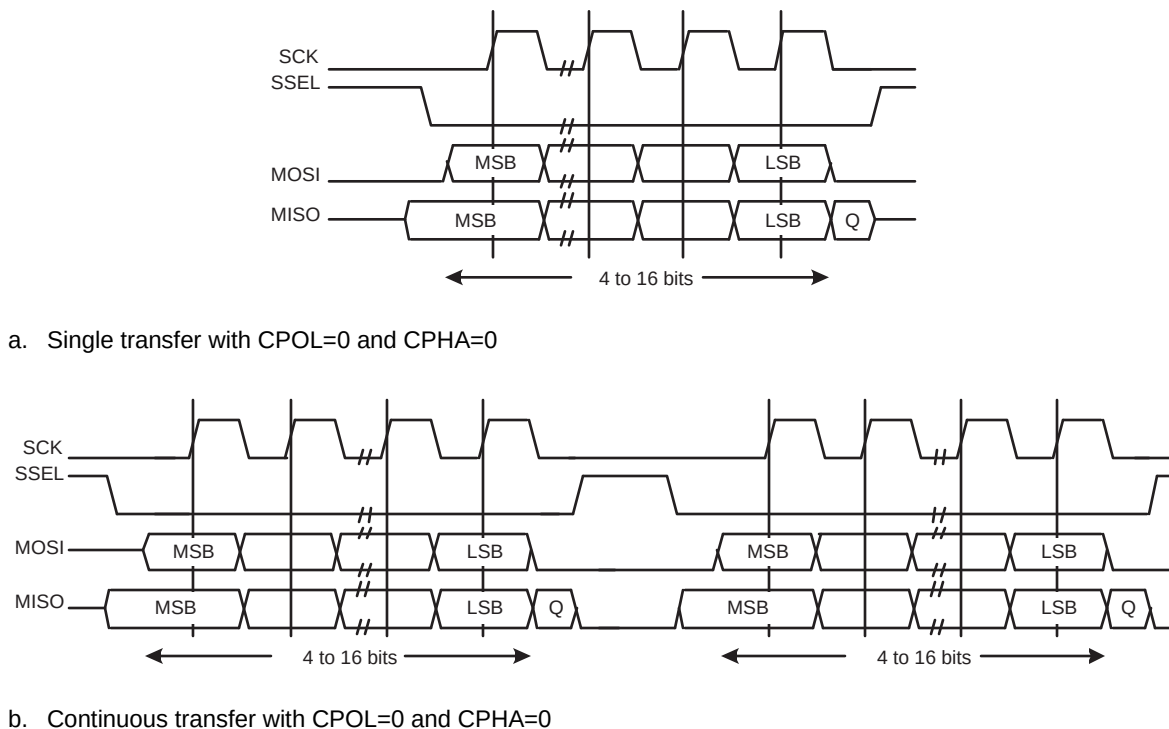


Fig 78. SPI frame format with CPOL=0 and CPHA=0 (a) Single and b) Continuous Transfer)

In this configuration, during idle periods:

- The CLK signal is forced LOW.
- SSEL is forced HIGH.
- The transmit MOSI/MISO pad is in high impedance.

If the SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSEL master signal being driven LOW. This causes slave data to be enabled onto the MISO input line of the master. Master's MOSI is enabled.

One half SCK period later, valid master data is transferred to the MOSI pin. Now that both the master and slave data have been set, the SCK master clock pin goes HIGH after one further half SCK period.

The data is now captured on the rising and propagated on the falling edges of the SCK signal.

In the case of a single word transmission, after all bits of the data word have been transferred, the SSEL line is returned to its idle HIGH state one SCK period after the last bit has been captured.

However, in the case of continuous back-to-back transmissions, the SSEL signal must be pulsed HIGH between each data word transfer. This is because the slave select pin freezes the data in its serial peripheral register and does not allow it to be altered if the CPHA bit is logic zero. Therefore the master device must raise the SSEL pin of the slave device between each data transfer to enable the serial peripheral data write. On completion of the continuous transfer, the SSEL pin is returned to its idle state one SCK period after the last bit has been captured.

18.5.2.3 SPI format with CPOL=0,CPHA=1

The transfer signal sequence for SPI format with CPOL = 0, CPHA = 1 is shown in [Figure 79](#), which covers both single and continuous transfers.

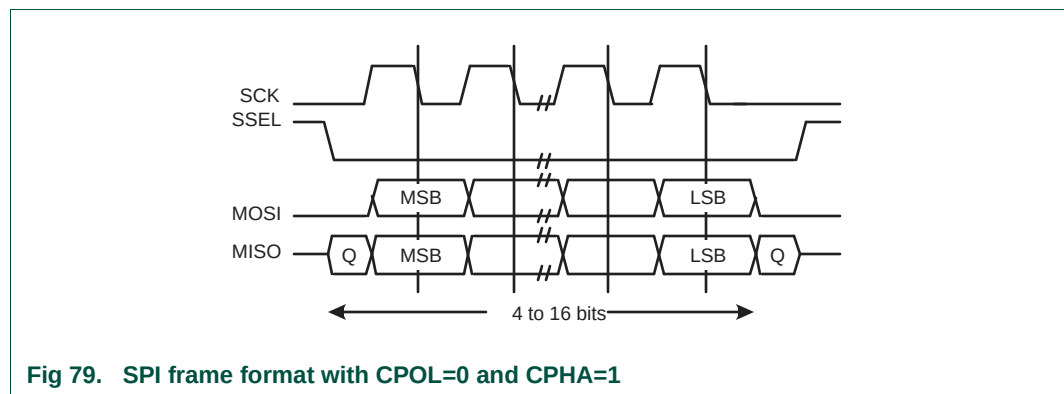


Fig 79. SPI frame format with CPOL=0 and CPHA=1

In this configuration, during idle periods:

- The CLK signal is forced LOW.
- SSEL is forced HIGH.
- The transmit MOSI/MISO pad is in high impedance.

If the SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSEL master signal being driven LOW. Master's MOSI pin is enabled. After a further one half SCK period, both master and slave valid data is enabled onto their respective transmission lines. At the same time, the SCK is enabled with a rising edge transition.

Data is then captured on the falling edges and propagated on the rising edges of the SCK signal.

In the case of a single word transfer, after all bits have been transferred, the SSEL line is returned to its idle HIGH state one SCK period after the last bit has been captured.

For continuous back-to-back transfers, the SSEL pin is held LOW between successive data words and termination is the same as that of the single word transfer.

18.5.2.4 SPI format with CPOL = 1,CPHA = 0

Single and continuous transmission signal sequences for SPI format with CPOL=1, CPHA=0 are shown in [Figure 80](#).

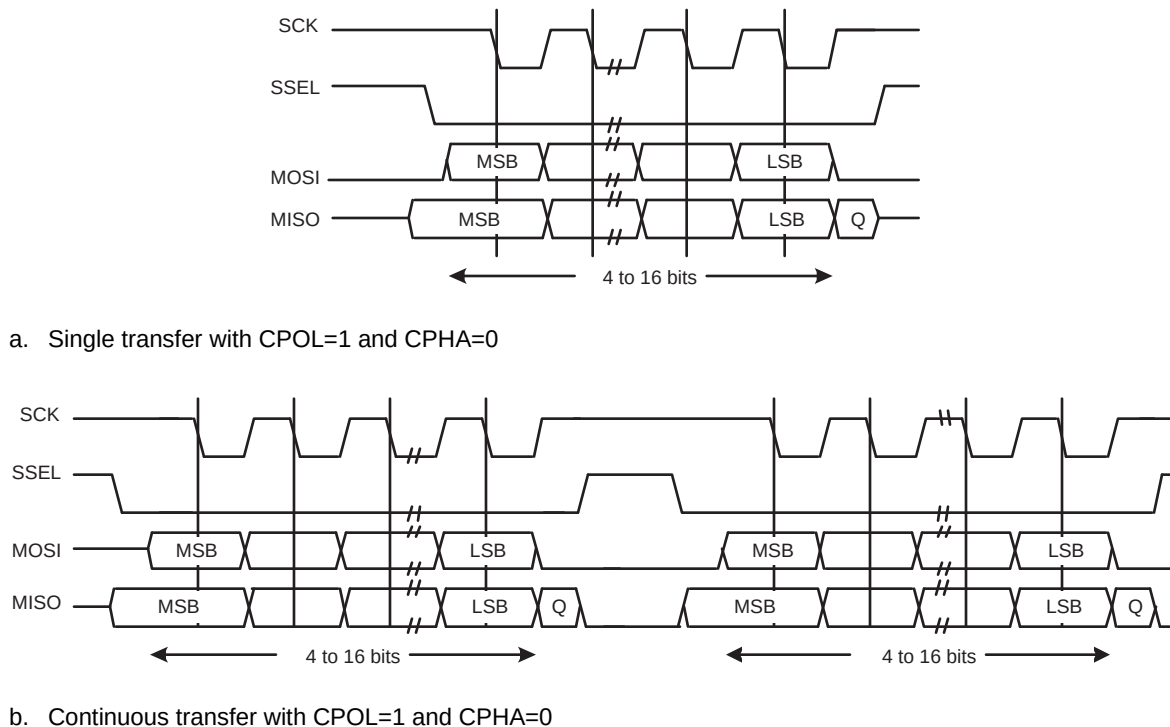


Fig 80. SPI frame format with CPOL = 1 and CPHA = 0 (a) Single and b) Continuous Transfer)

In this configuration, during idle periods:

- The CLK signal is forced HIGH.
- SSEL is forced HIGH.
- The transmit MOSI/MISO pad is in high impedance.

If the SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSEL master signal being driven LOW, which causes slave data to be immediately transferred onto the MISO line of the master. Master's MOSI pin is enabled.

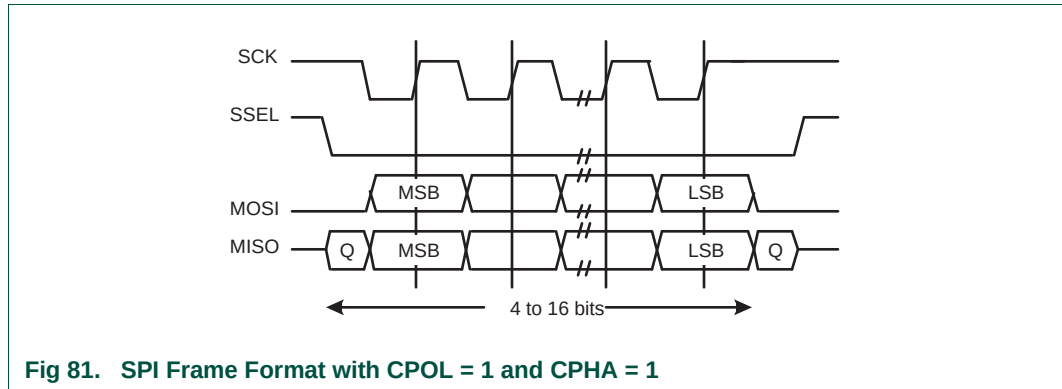
One half period later, valid master data is transferred to the MOSI line. Now that both the master and slave data have been set, the SCK master clock pin becomes LOW after one further half SCK period. This means that data is captured on the falling edges and be propagated on the rising edges of the SCK signal.

In the case of a single word transmission, after all bits of the data word are transferred, the SSEL line is returned to its idle HIGH state one SCK period after the last bit has been captured.

However, in the case of continuous back-to-back transmissions, the SSEL signal must be pulsed HIGH between each data word transfer. This is because the slave select pin freezes the data in its serial peripheral register and does not allow it to be altered if the CPHA bit is logic zero. Therefore the master device must raise the SSEL pin of the slave device between each data transfer to enable the serial peripheral data write. On completion of the continuous transfer, the SSEL pin is returned to its idle state one SCK period after the last bit has been captured.

18.5.2.5 SPI format with CPOL = 1, CPHA = 1

The transfer signal sequence for SPI format with CPOL = 1, CPHA = 1 is shown in [Figure 81](#), which covers both single and continuous transfers.



In this configuration, during idle periods:

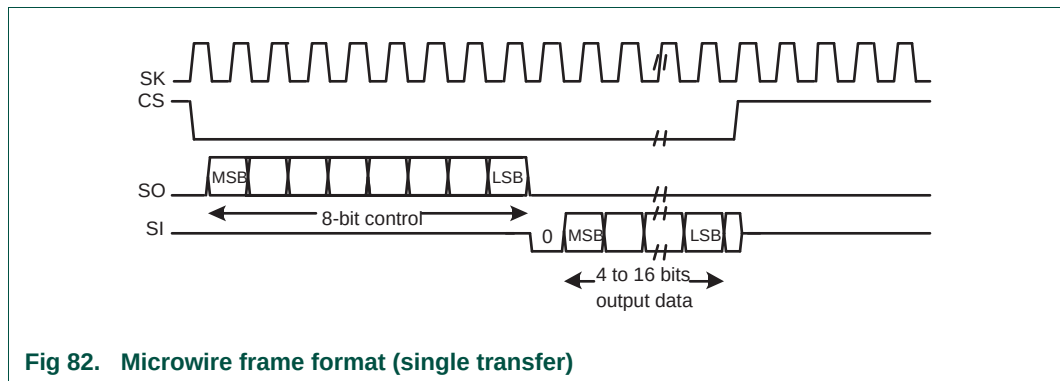
- The CLK signal is forced HIGH.
- SSEL is forced HIGH.
- The transmit MOSI/MISO pad is in high impedance.

If the SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSEL master signal being driven LOW. Master's MOSI is enabled. After a further one half SCK period, both master and slave data are enabled onto their respective transmission lines. At the same time, the SCK is enabled with a falling edge transition. Data is then captured on the rising edges and propagated on the falling edges of the SCK signal.

After all bits have been transferred, in the case of a single word transmission, the SSEL line is returned to its idle HIGH state one SCK period after the last bit has been captured. For continuous back-to-back transmissions, the SSEL pins remains in its active LOW state, until the final bit of the last word has been captured, and then returns to its idle state as described above. In general, for continuous back-to-back transfers the SSEL pin is held LOW between successive data words and termination is the same as that of the single word transfer.

18.5.3 National Semiconductor Microwire frame format

[Figure 82](#) shows the Microwire frame format for a single frame. [Figure 83](#) shows the same format when back-to-back frames are transmitted.



Microwire format is very similar to SPI format, except that transmission is half-duplex instead of full-duplex, using a master-slave message passing technique. Each serial transmission begins with an 8-bit control word that is transmitted from the SSP to the off-chip slave device. During this transmission, no incoming data is received by the SSP. After the message has been sent, the off-chip slave decodes it and, after waiting one serial clock after the last bit of the 8-bit control message has been sent, responds with the required data. The returned data is 4 to 16 bits in length, making the total frame length anywhere from 13 to 25 bits.

In this configuration, during idle periods:

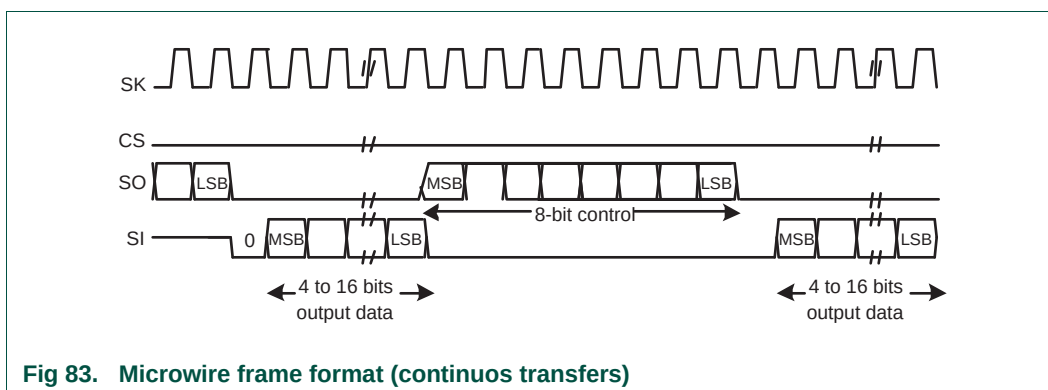
- The SK signal is forced LOW.
- CS is forced HIGH.
- The transmit data line SO is arbitrarily forced LOW.

A transmission is triggered by writing a control byte to the transmit FIFO. The falling edge of CS causes the value contained in the bottom entry of the transmit FIFO to be transferred to the serial shift register of the transmit logic, and the MSB of the 8-bit control frame to be shifted out onto the SO pin. CS remains LOW for the duration of the frame transmission. The SI pin remains tristated during this transmission.

The off-chip serial slave device latches each control bit into its serial shifter on the rising edge of each SK. After the last bit is latched by the slave device, the control byte is decoded during a one clock wait-state, and the slave responds by transmitting data back to the SSP. Each bit is driven onto SI line on the falling edge of SK. The SSP in turn latches each bit on the rising edge of SK. At the end of the frame, for single transfers, the CS signal is pulled HIGH one clock period after the last bit has been latched in the receive serial shifter, that causes the data to be transferred to the receive FIFO.

Note: The off-chip slave device can tristate the receive line either on the falling edge of SK after the LSB has been latched by the receive shifter, or when the CS pin goes HIGH.

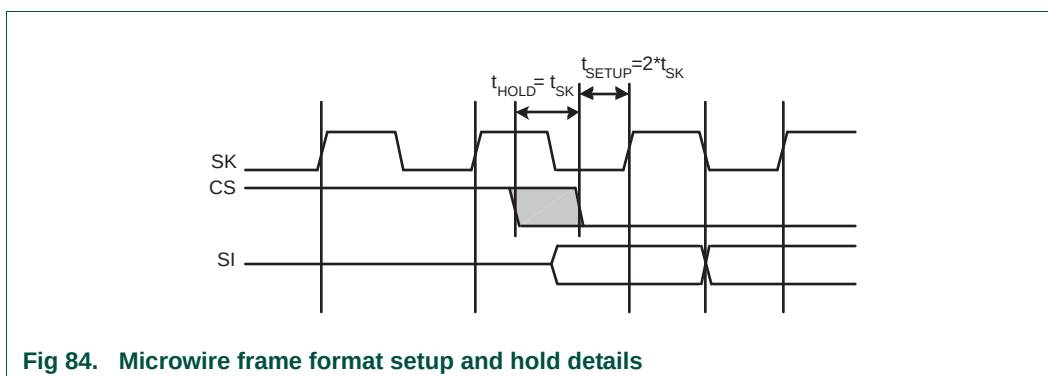
For continuous transfers, data transmission begins and ends in the same manner as a single transfer. However, the CS line is continuously asserted (held LOW) and transmission of data occurs back to back. The control byte of the next frame follows directly after the LSB of the received data from the current frame. Each of the received values is transferred from the receive shifter on the falling edge SK, after the LSB of the frame has been latched into the SSP.



18.5.3.1 Setup and hold time requirements on CS with respect to SK in Microwire mode

In the Microwire mode, the SSP slave samples the first bit of receive data on the rising edge of SK after CS has gone LOW. Masters that drive a free-running SK must ensure that the CS signal has sufficient setup and hold margins with respect to the rising edge of SK.

[Figure 84](#) illustrates these setup and hold time requirements. With respect to the SK rising edge on which the first bit of receive data is to be sampled by the SSP slave, CS must have a setup of at least two times the period of SK on which the SSP operates. With respect to the SK rising edge previous to this edge, CS must have a hold of at least one SK period.



18.6 Register description

The register addresses of the SSP controllers addresses are shown in [Table 370](#).

Table 370. SSP Register Map

Generic Name	Description	Access	Reset Value ^[1]	SSPn Register Name & Address
CR0	Control Register 0. Selects the serial clock rate, bus type, and data size.	R/W	0	SSP0CR0 - 0x4008 8000 SSP1CR0 - 0x4003 0000
CR1	Control Register 1. Selects master/slave and other modes.	R/W	0	SSP0CR1 - 0x4008 8004 SSP1CR1 - 0x4003 0004
DR	Data Register. Writes fill the transmit FIFO, and reads empty the receive FIFO.	R/W	0	SSP0DR - 0x4008 8008 SSP1DR - 0x4003 0008
SR	Status Register	RO		SSP0SR - 0x4008 800C SSP1SR - 0x4003 000C
CPSR	Clock Prescale Register	R/W	0	SSP0CPSR - 0x4008 8010 SSP1CPSR - 0x4003 0010
IMSC	Interrupt Mask Set and Clear Register	R/W	0	SSP0IMSC - 0x4008 8014 SSP1IMSC - 0x4003 0014
RIS	Raw Interrupt Status Register	R/W		SSP0RIS - 0x4008 8018 SSP1RIS - 0x4003 0018
MIS	Masked Interrupt Status Register	R/W	0	SSP0MIS - 0x4008 801C SSP1MIS - 0x4003 001C
ICR	SSPICR Interrupt Clear Register	R/W	NA	SSP0ICR - 0x4008 8020 SSP1ICR - 0x4003 0020
DMACR	DMA Control Register	R/W	0	SSP0DMACR - 0x4008 8024 SSP1DMACR - 0x4003 0024

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

18.6.1 SSPn Control Register 0 (SSP0CR0 - 0x4008 8000, SSP1CR0 - 0x4003 0000)

This register controls the basic operation of the SSP controller.

Table 371: SSPn Control Register 0 (SSP0CR0 - address 0x4008 8000, SSP1CR0 - 0x4003 0000) bit description

Bit	Symbol	Value	Description	Reset Value
3:0	DSS		Data Size Select. This field controls the number of bits transferred in each frame. Values 0000-0010 are not supported and should not be used.	0000
		0011	4-bit transfer	
		0100	5-bit transfer	
		0101	6-bit transfer	
		0110	7-bit transfer	
		0111	8-bit transfer	
		1000	9-bit transfer	
		1001	10-bit transfer	
		1010	11-bit transfer	
		1011	12-bit transfer	
		1100	13-bit transfer	
		1101	14-bit transfer	
		1110	15-bit transfer	
		1111	16-bit transfer	
5:4	FRF		Frame Format.	00
		00	SPI	
		01	TI	
		10	Microwire	
		11	This combination is not supported and should not be used.	
6	CPOL		Clock Out Polarity. This bit is only used in SPI mode.	0
		0	SSP controller maintains the bus clock low between frames.	
		1	SSP controller maintains the bus clock high between frames.	
7	CPHA		Clock Out Phase. This bit is only used in SPI mode.	0
		0	SSP controller captures serial data on the first clock transition of the frame, that is, the transition away from the inter-frame state of the clock line.	
		1	SSP controller captures serial data on the second clock transition of the frame, that is, the transition back to the inter-frame state of the clock line.	
15:8	SCR		Serial Clock Rate. The number of prescaler-output clocks per bit on the bus, minus one. Given that CPSDVSR is the prescale divider, and the APB clock PCLK clocks the prescaler, the bit frequency is $PCLK / (CPSDVSR \times [SCR+1])$.	0x00
31:8	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.2 SSPn Control Register 1 (SSP0CR1 - 0x4008 8004, SSP1CR1 - 0x4003 0004)

This register controls certain aspects of the operation of the SSP controller.

Table 372: SSPn Control Register 1 (SSP0CR1 - address 0x4008 8004, SSP1CR1 - 0x4003 0004) bit description

Bit	Symbol	Value	Description	Reset Value
0	LBM		Loop Back Mode.	0
		0	During normal operation.	
		1	Serial input is taken from the serial output (MOSI or MISO) rather than the serial input pin (MISO or MOSI respectively).	
1	SSE		SSP Enable.	0
		0	The SSP controller is disabled.	
		1	The SSP controller will interact with other devices on the serial bus. Software should write the appropriate control information to the other SSP registers and interrupt controller registers, before setting this bit.	
2	MS		Master/Slave Mode. This bit can only be written when the SSE bit is 0.	0
		0	The SSP controller acts as a master on the bus, driving the SCLK, MOSI, and SSEL lines and receiving the MISO line.	
		1	The SSP controller acts as a slave on the bus, driving MISO line and receiving SCLK, MOSI, and SSEL lines.	
3	SOD		Slave Output Disable. This bit is relevant only in slave mode (MS = 1). If it is 1, this blocks this SSP controller from driving the transmit data line (MISO).	0
31:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.3 SSPn Data Register (SSP0DR - 0x4008 8008, SSP1DR - 0x4003 0008)

Software can write data to be transmitted to this register, and read data that has been received.

Table 373: SSPn Data Register (SSP0DR - address 0x4008 8008, SSP1DR - 0x4003 0008) bit description

Bit	Symbol	Description	Reset Value
15:0	DATA	Write: software can write data to be sent in a future frame to this register whenever the TNF bit in the Status register is 1, indicating that the Tx FIFO is not full. If the Tx FIFO was previously empty and the SSP controller is not busy on the bus, transmission of the data will begin immediately. Otherwise the data written to this register will be sent as soon as all previous data has been sent (and received). If the data length is less than 16 bits, software must right-justify the data written to this register. Read: software can read data from this register whenever the RNE bit in the Status register is 1, indicating that the Rx FIFO is not empty. When software reads this register, the SSP controller returns data from the least recent frame in the Rx FIFO. If the data length is less than 16 bits, the data is right-justified in this field with higher order bits filled with 0s.	0x0000
31:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.4 SSPn Status Register (SSP0SR - 0x4008 800C, SSP1SR - 0x4003 000C)

This read-only register reflects the current status of the SSP controller.

Table 374: SSPn Status Register (SSP0SR - address 0x4008 800C, SSP1SR - 0x4003 000C) bit description

Bit	Symbol	Description	Reset Value
0	TFE	Transmit FIFO Empty. This bit is 1 if the Transmit FIFO is empty, 0 if not.	1
1	TNF	Transmit FIFO Not Full. This bit is 0 if the Tx FIFO is full, 1 if not.	1
2	RNE	Receive FIFO Not Empty. This bit is 0 if the Receive FIFO is empty, 1 if not.	0
3	RFF	Receive FIFO Full. This bit is 1 if the Receive FIFO is full, 0 if not.	0
4	BSY	Busy. This bit is 0 if the SSPn controller is idle, or 1 if it is currently sending/receiving a frame and/or the Tx FIFO is not empty.	0
31:5	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.5 SSPn Clock Prescale Register (SSP0CPSR - 0x4008 8010, SSP1CPSR - 0x4003 0010)

This register controls the factor by which the Prescaler divides the SSP peripheral clock SSP_PCLK to yield the prescaler clock that is, in turn, divided by the SCR factor in SSPnCR0, to determine the bit clock.

Table 375: SSPn Clock Prescale Register (SSP0CPSR - address 0x4008 8010, SSP1CPSR - 0x4003 0010) bit description

Bit	Symbol	Description	Reset Value
7:0	CPSDVSR	This even value between 2 and 254, by which SSP_PCLK is divided to yield the prescaler output clock. Bit 0 always reads as 0.	0
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Important: the SSPnCPSR value must be properly initialized or the SSP controller will not be able to transmit data correctly.

In Slave mode, the SSP clock rate provided by the master must not exceed 1/12 of the SSP peripheral clock selected in [Section 4.7.3](#). The content of the SSPnCPSR register is not relevant.

In master mode, $CPSDVSR_{min} = 2$ or larger (even numbers only).

18.6.6 SSPn Interrupt Mask Set/Clear Register (SSP0IMSC - 0x4008 8014, SSP1IMSC - 0x4003 0014)

This register controls whether each of the four possible interrupt conditions in the SSP controller are enabled. Note that ARM uses the word “masked” in the opposite sense from classic computer terminology, in which “masked” meant “disabled”. ARM uses the word “masked” to mean “enabled”. To avoid confusion we will not use the word “masked”.

Table 376: SSPn Interrupt Mask Set/Clear register (SSP0IMSC - address 0x4008 8014, SSP1IMSC - 0x4003 0014) bit description

Bit	Symbol	Description	Reset Value
0	RORIM	Software should set this bit to enable interrupt when a Receive Overrun occurs, that is, when the Rx FIFO is full and another frame is completely received. The ARM spec implies that the preceding frame data is overwritten by the new frame data when this occurs.	0
1	RTIM	Software should set this bit to enable interrupt when a Receive Time-out condition occurs. A Receive Time-out occurs when the Rx FIFO is not empty, and no has not been read for a "time-out period". The time-out period is the same for master and slave modes and is determined by the SSP bit rate: 32 bits at PCLK / (CPSDVSr × [SCR+1]).	0
2	RXIM	Software should set this bit to enable interrupt when the Rx FIFO is at least half full.	0
3	TXIM	Software should set this bit to enable interrupt when the Tx FIFO is at least half empty.	0
31:4	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.7 SSPn Raw Interrupt Status Register (SSP0RIS - 0x4008 8018, SSP1RIS - 0x4003 0018)

This read-only register contains a 1 for each interrupt condition that is asserted, regardless of whether or not the interrupt is enabled in the SSPnIMSC.

Table 377: SSPn Raw Interrupt Status register (SSP0RIS - address 0x4008 8018, SSP1RIS - 0x4003 0018) bit description

Bit	Symbol	Description	Reset Value
0	RORRIS	This bit is 1 if another frame was completely received while the Rx FIFO was full. The ARM spec implies that the preceding frame data is overwritten by the new frame data when this occurs.	0
1	RTRIS	This bit is 1 if the Rx FIFO is not empty, and has not been read for a "time-out period". The time-out period is the same for master and slave modes and is determined by the SSP bit rate: 32 bits at PCLK / (CPSDVSr × [SCR+1]).	0
2	RXRIS	This bit is 1 if the Rx FIFO is at least half full.	0
3	TXRIS	This bit is 1 if the Tx FIFO is at least half empty.	1
31:4	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.8 SSPn Masked Interrupt Status Register (SSP0MIS - 0x4008 801C, SSP1MIS - 0x4003 001C)

This read-only register contains a 1 for each interrupt condition that is asserted and enabled in the SSPnIMSC. When an SSP interrupt occurs, the interrupt service routine should read this register to determine the cause(s) of the interrupt.

Table 378: SSPn Masked Interrupt Status register (SSPnMIS -address 0x4008 801C, SSP1MIS - 0x4003 001C) bit description

Bit	Symbol	Description	Reset Value
0	RORMIS	This bit is 1 if another frame was completely received while the RxFIFO was full, and this interrupt is enabled.	0
1	RTMIS	This bit is 1 if the Rx FIFO is not empty, has not been read for a "time-out period", and this interrupt is enabled. The time-out period is the same for master and slave modes and is determined by the SSP bit rate: 32 bits at $PCLK / (CPSDVSR \times [SCR+1])$.	0
2	RXMIS	This bit is 1 if the Rx FIFO is at least half full, and this interrupt is enabled.	0
3	TXMIS	This bit is 1 if the Tx FIFO is at least half empty, and this interrupt is enabled.	0
31:4	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.9 SSPn Interrupt Clear Register (SSP0ICR - 0x4008 8020, SSP1ICR - 0x4003 0020)

Software can write one or more one(s) to this write-only register, to clear the corresponding interrupt condition(s) in the SSP controller. Note that the other two interrupt conditions can be cleared by writing or reading the appropriate FIFO, or disabled by clearing the corresponding bit in SSPnIMSC.

Table 379: SSPn interrupt Clear Register (SSP0ICR - address 0x4008 8020, SSP1ICR - 0x4003 0020) bit description

Bit	Symbol	Description	Reset Value
0	RORIC	Writing a 1 to this bit clears the "frame was received when RxFIFO was full" interrupt.	NA
1	RTIC	Writing a 1 to this bit clears the "Rx FIFO was not empty and has not been read for a time-out period" interrupt. The time-out period is the same for master and slave modes and is determined by the SSP bit rate: 32 bits at $PCLK / (CPSDVSR \times [SCR+1])$.	NA
31:2	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

18.6.10 SSPn DMA Control Register (SSP0DMACR - 0x4008 8024, SSP1DMACR - 0x4003 0024)

The SSPnDMACR register is the DMA control register. It is a read/write register.

Table 380: SSPn DMA Control Register (SSP0DMACR - address 0x4008 8024, SSP1DMACR - 0x4003 0024) bit description

Bit	Symbol	Description	Reset Value
0	Receive DMA Enable (RXDMAE)	When this bit is set to one 1, DMA for the receive FIFO is enabled, otherwise receive DMA is disabled.	0
1	Transmit DMA Enable (TXDMAE)	When this bit is set to one 1, DMA for the transmit FIFO is enabled, otherwise transmit DMA is disabled	0
31:2	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA